

nssDMRG: an Entanglement-Informed Distributed DMRG Method

Method report

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Abstract

nssDMRG is a distributed-memory implementation of the density-matrix renormalization group for one-dimensional interacting spin and fermionic systems. Its computational design follows directly from the bipartite and symmetry-resolved structure of the many-body wavefunction. At every DMRG step, the superblock state is represented as a collection of matrices indexed by conserved Abelian quantum numbers. Distributing the columns of these matrices across MPI processes makes one class of Hamiltonian contractions strictly local, while the complementary contractions are evaluated using communication-efficient distributed transpositions. The full superblock Hamiltonian need not be assembled: its action on a vector is reduced to sector-restricted block operations that form the kernel of a Krylov eigensolver. This report summarizes the mathematical construction, the DMRG workflow, and the main computational choices—sparse versus direct application, cached versus lazy filtering, symmetry-aware communicators, boundary conditions, and persistent storage—without entering the details of individual source-code procedures.

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1 Overview

The density-matrix renormalization group (DMRG) is an adaptive variational method for quantum lattice systems. Its efficiency in one dimension follows from the fact that physically relevant low-energy states often have limited bipartite entanglement. Instead of retaining a fixed local basis as the system grows, DMRG repeatedly selects the states carrying the dominant Schmidt weight across a cut.

The same entanglement cut also provides a natural distributed representation of the wavefunction. This observation is the organizing principle of **nssDMRG**: the computational data layout is not imposed independently of the many-body state, but is inherited from its product-basis and symmetry structure. The representation serves two purposes simultaneously:

- (i) it identifies the optimal retained states through the reduced density matrix;
- (ii) it decomposes the dominant Hamiltonian–vector product into local contractions and a small number of structured collective communications.

The library supports infinite- and finite-system DMRG, Abelian quantum-number conservation, spin and Hubbard-type fermionic models, open and periodic boundary conditions, MPI-distributed superblock diagonalization, restart and checkpoint facilities, and post-processing measurements based on stored renormalization data.

2 Lattice problem and symmetry decomposition

Consider a finite lattice with Hilbert space

$$\mathcal{H} = \bigotimes_{i=1}^S \mathcal{H}_i \quad (1)$$

and a finite-range, or more generally K -local, Hamiltonian

$$H = \sum_{n=1}^K \sum_{i_1, \dots, i_n} h_{i_1, \dots, i_n}^{(n)}, \quad h_{i_1, \dots, i_n}^{(n)} \propto \prod_{m=1}^n O_{i_m}. \quad (2)$$

An entanglement cut partitions the lattice into left and right components,

$$\mathcal{H} = \mathcal{H}_L \otimes \mathcal{H}_R. \quad (3)$$

We assume an Abelian conserved quantity, or a set of mutually commuting quantities, denoted collectively by Q :

$$[H, Q] = 0. \quad (4)$$

The partition is chosen so that each subsystem carries well-defined quantum numbers. For additive symmetries,

$$q_L + q_R = Q; \quad (5)$$

more generally, one may write $q_L \oplus q_R = Q$, with $\oplus$ the Abelian composition law. Typical examples are

$$q = S^z \quad (6)$$

for spin chains and

$$\mathbf{q} = (N_\uparrow, N_\downarrow) \quad (7)$$

for number-conserving electronic systems.

Within a fixed total sector Q , only compatible pairs (q_L, q_R) occur. The wavefunction decomposes as

$$|\Psi\rangle = \sum_q \sum_{l_q=1}^{d_q^L} \sum_{r_q=1}^{d_q^R} \Psi_{r_q l_q}^q |l_q\rangle_L \otimes |r_q\rangle_R. \quad (8)$$

Equation (??) establishes an isomorphism between a symmetry-resolved state vector and a collection of dense matrices

$$\hat{\Psi}^q \in \mathbb{C}^{d_q^R \times d_q^L}. \quad (9)$$

This matrix representation is central both to entanglement truncation and to the parallel Hamiltonian application.

3 DMRG renormalization cycle

3.1 Block enlargement and superblock construction

At a generic step, renormalized left and right blocks are enlarged by one or more physical sites. For open boundary conditions, the left block grows toward the right and the right block grows toward the left. The enlarged blocks form a superblock,

$$\mathcal{H}_{\text{SB}} = \mathcal{H}_L \otimes \mathcal{H}_R, \quad (10)$$

restricted to the target total quantum-number sector.

The superblock Hamiltonian has the generic bipartite form

$$H_{\text{SB}} = H_L \otimes \mathbb{1}_R + \mathbb{1}_L \otimes H_R + \sum_m A_m \otimes B_m. \quad (11)$$

The first two terms contain the Hamiltonians internal to each block. The finite set of products $A_m \otimes B_m$ contains interactions crossing the cut: spin couplings, hopping processes, density interactions, or other model-dependent links.

In a symmetry-adapted basis, every object in Eq. (??) is resolved into blocks. A local operator O carries a quantum-number shift

$$dq(O) = q_{\text{out}} - q_{\text{in}}, \quad O : \mathcal{H}(q_{\text{in}}) \longrightarrow \mathcal{H}(q_{\text{out}}). \quad (12)$$

For each coupling $A_m \otimes B_m$, total symmetry conservation requires

$$dq(A_m) + dq(B_m) = 0. \quad (13)$$

This metadata determines exactly which rectangular symmetry blocks can contribute, without relying on model-specific assumptions.

3.2 Ground-state optimization

The lowest state of H_{SB} is found with an iterative Krylov-space eigensolver. Such algorithms require repeated evaluations of

$$|y\rangle = H_{\text{SB}}|x\rangle, \quad (14)$$

but do not intrinsically require storage of the full Hamiltonian. Since this matrix–vector product dominates the run time of large calculations, its representation and parallel execution determine the practical scalability of the method.

3.3 Reduced density matrix and truncation

Once the optimized state is available, the left reduced density matrix is

$$\rho_L = \text{Tr}_R |\Psi\rangle\langle\Psi|. \quad (15)$$

Using Eq. (??), it is block diagonal:

$$\rho_L = \bigoplus_q \rho_L^q, \quad \rho_L^q = (\hat{\Psi}^q)^\dagger \hat{\Psi}^q. \quad (16)$$

An analogous expression holds for ρ_R . Diagonalizing these matrices gives the Schmidt probabilities $\lambda_{q\alpha}$ and the corresponding renormalization transformations. The retained basis is selected either by a maximum bond dimension χ or by a discarded-weight criterion. Its sector-resolved dimensions obey

$$\sum_q \chi_q^L = \chi_L, \quad \sum_q \chi_q^R = \chi_R. \quad (17)$$

For a retained isometry U , every block operator is transformed according to

$$O \longmapsto U^\dagger O U. \quad (18)$$

The renormalized blocks are then reused in the next growth step or in the next finite-system sweep. The truncation error is the total discarded density matrix weight,

$$\epsilon_{\text{tr}} = 1 - \sum_{(q,\alpha) \in \text{kept}} \lambda_{q\alpha}. \quad (19)$$

4 Direct Hamiltonian application

4.1 Vectorization identity

Let a superblock sector be represented by the matrix X^q associated with a vector $|x_q\rangle = \text{Vec}(X^q)$, where columns are stacked. For compatible matrices A and B ,

$$(A \otimes B) \text{Vec}(X) = \text{Vec}(BXA^\top). \quad (20)$$

This identity replaces a very large sparse Kronecker product by two much smaller matrix contractions.

Applied to Eq. (??), it yields

$$(\mathbb{K}_L \otimes H_R) \text{Vec}(X) = \text{Vec}(H_R X), \quad (21)$$

$$(H_L \otimes \mathbb{K}_R) \text{Vec}(X) = \text{Vec}(X H_L^\top), \quad (22)$$

$$(A_m \otimes B_m) \text{Vec}(X) = \text{Vec}(B_m X A_m^\top). \quad (23)$$

The superblock Hamiltonian is therefore applied term by term, exploiting its tensor-product and symmetry-block structure. The direct formulation avoids assembling H_{SB} , whose dimension grows as the product of the two block dimensions.

4.2 Sector-restricted products

Suppose that a coupling operator connects an input left sector k to an output sector q . Its filtered block is

$$A_m(q, k) : \mathcal{H}_L(k) \longrightarrow \mathcal{H}_L(q), \quad (24)$$

and the right factor performs the complementary transition dictated by Eq. (??). The corresponding contribution is

$$\text{Vec} \left[B_m(q, k) X^k A_m(q, k)^\top \right]. \quad (25)$$

Only symmetry-compatible pairs (q, k) are evaluated. This substantially reduces both arithmetic and storage compared with operations in the full product basis.

5 Entanglement-informed MPI distribution

5.1 Column-wise ownership

For each symmetry block $X^q \in \mathbb{C}^{d_q^R \times d_q^L}$, columns are distributed among P MPI processes. Process p owns a subset of the left basis states and the associated column panel

$$X_p^q \in \mathbb{C}^{d_q^R \times \Theta_{q,p}}, \quad \sum_p \Theta_{q,p} = d_q^L. \quad (26)$$

This decomposition is entanglement-informed because the matrix blocks are exactly those induced by the symmetry-preserving entanglement cut. In DMRG, their dimensions are the sector-resolved bond dimensions selected from the Schmidt spectrum.

The layout has an immediate computational advantage. Left multiplication by a right-block operator is local:

$$Y_p^q = H_R^q X_p^q. \quad (27)$$

Every process possesses the complete row extent needed for each of its local columns, and no communication is required.

5.2 Distributed transposition

Right multiplication by a left-block operator couples columns that reside on different processes. Instead of gathering the full block, the method applies a distributed transposition, denoted by T^* ,

$$\{X_p^q\}_{p=1}^P \xrightarrow{T^*} \{(X^q)^\top\}_{p=1}^P. \quad (28)$$

It is implemented through an all-to-all variable-size data exchange. After the transposition, the formerly nonlocal contraction becomes local; a second T^* restores the original distribution:

$$X^q \xrightarrow{T^*} (X^q)^\top \xrightarrow{H_L^q} H_L^q (X^q)^\top \xrightarrow{T^*} X^q (H_L^q)^\top. \quad (29)$$

5.3 Coupling terms

For Eq. (??), the right factor is applied first:

$$C_m(q, k) = B_m(q, k) X^k. \quad (30)$$

This multiplication is local and the intermediate matrix inherits the column-wise distribution of X^k . The remaining contraction is rewritten as

$$C_m(q, k) A_m(q, k)^\top = \left[A_m(q, k) C_m(q, k)^{\top*} \right]^{T^*}. \quad (31)$$

It therefore requires two distributed transpositions but no global assembly of either the wavefunction or the Kronecker operator. The complete Hamiltonian application consists of local sparse-dense contractions plus a small, structured set of collective communications.

6 Entanglement, workload, and symmetry fragmentation

The entanglement Hamiltonian and entropy associated with the cut are

$$H_E = -\log \rho_L, \quad S_L = -\text{Tr}(\rho_L \log \rho_L). \quad (32)$$

Writing the entanglement levels as $\xi_{q\alpha} = -\log \lambda_{q\alpha}$, the weight of sector q is

$$W_q = \frac{\sum_{\alpha} e^{-\xi_{q\alpha}}}{\sum_{k, \alpha} e^{-\xi_{k\alpha}}}. \quad (33)$$

Since columns of $\hat{\Psi}^q$ are distributed across processes, the same sector structure that carries the Schmidt weight also determines the distributed workload. A process-level weight may be defined as

$$w_p = \sum_{q \in \mathcal{Q}_p} W_q, \quad \mathcal{Q}_p = \{q : \Theta_{q,p} > 0\}, \quad (34)$$

with inverse participation ratio

$$W_P = \sum_{p=1}^P w_p^2. \quad (35)$$

This quantity characterizes how broadly the entanglement weight is resolved over the process-owned subspaces.

The sector dimensions are generally nonuniform. A few large sectors coexist with many smaller sectors, producing *symmetry fragmentation*. This fragmentation controls load balance and the ratio between communication and computation. Arithmetic is governed primarily by the volume of the matrix blocks, while distributed transposition is controlled by their communicated surface. Consequently, increasing the total bond dimension improves strong scaling until local panels become too small and latency dominates.

6.1 Symmetry-aware subcommunicators

A sector with dimensions $d_q^L \times d_q^R$ cannot usefully employ more processes than its available rows or columns. Assigning all global processes to every small sector would create idle ranks and unnecessary collective communication. The implementation therefore uses a sector-dependent active process count of the form

$$P_{\text{active}}(q) = \max\left[\min(d_q^L, P), \min(d_q^R, P)\right], \quad (36)$$

and constructs MPI subcommunicators accordingly. Each sector is processed only by ranks that own a nonzero share in at least one relevant layout. This reduces symmetry-induced imbalance and prevents small blocks from imposing global synchronization on the full process pool.

7 Computational modes

The superblock eigensolver can use three principal Hamiltonian representations.

Mode	Persistent objects	Main trade-off
Sparse	Explicit sparse H_{SB}	Simple application, but potentially prohibitive superblock memory.
Direct cached	Sector-filtered $H_L(q)$, $H_R(q)$, $A_m(q, k)$, and $B_m(q, k)$	Fast repeated Krylov products at the cost of a larger persistent footprint.
Direct lazy	Full enlarged block operators, sector state lists, and inverse maps	Smaller persistent memory; filtered blocks are rebuilt during each application.

Table 1: High-level Hamiltonian-application strategies.

The sparse mode explicitly constructs the superblock Hamiltonian. It is useful for smaller problems and as an independent reference path. The direct modes exploit Eq. (??) and are preferable when explicit superblock storage becomes limiting.

In cached direct mode, all symmetry-filtered operator blocks needed by the Krylov iteration are prepared once and reused. This minimizes filtering overhead but may consume memory comparable to an additional distributed copy of the enlarged operator content.

In lazy direct mode, only the enlarged operators and lightweight maps between full and sector-local indices remain resident. A required block

$$O(q, k) = O|_{\mathcal{H}(k) \rightarrow \mathcal{H}(q)} \quad (37)$$

is constructed immediately before use and released afterwards. The mode trades repeated sparse filtering for a substantially smaller memory footprint, which permits larger retained bases in memory-bound calculations. Cached and lazy modes share the same mathematical Hamiltonian action and MPI data layout.

8 Spin and fermionic models

For a spin chain, a representative nearest-neighbor Hamiltonian is

$$H = J_z \sum_i S_i^z S_{i+1}^z + \frac{J_{xy}}{2} \sum_i \left(S_i^+ S_{i+1}^- + S_i^- S_{i+1}^+ \right). \quad (38)$$

The connecting terms naturally take the form $A_m \otimes B_m$, and their quantum-number shifts cancel pairwise.

For a Hubbard chain,

$$H = - \sum_{i,\sigma} t_{i\sigma} \left(c_{i\sigma}^\dagger c_{i+1,\sigma} + \text{H.c.} \right) + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad (39)$$

fermionic anticommutation must be respected across tensor-product boundaries. With the convention used by the code, a boundary hopping factor on the left is represented using the parity operator $P = (-1)^N$, for example

$$(c_{L\sigma}^\dagger P_L) \otimes c_{R\sigma}. \quad (40)$$

The parity factor is prepared consistently in both cached and lazy direct modes. Its presence changes the algebraic operator content but not the distributed contraction pattern.

9 Open and periodic boundary conditions

With open boundaries, each block grows toward the central cut and the superblock contains a single set of links connecting the enlarged blocks. Periodic boundaries add couplings closing the two outer ends. A naive growth scheme would leave one boundary operator buried under many renormalization steps, degrading its representation.

The periodic strategy therefore alternates block growth on the two sides. Each block carries operators associated with both link directions, and the superblock is connected at both ends. This preserves a more balanced renormalization history for the two periodic links. The physical ordering of sites is tracked separately from their growth order, a distinction that is also required when reconstructing nonlocal observables.

10 Memory management, restart, and measurements

Large DMRG calculations must control persistent storage as carefully as the Hamiltonian application. The implementation separates several classes of data:

- renormalized block objects required to continue or restart a run;
- rotation matrices U_n required by finite sweeps and by post-processing operators;
- stored superblock states and sector metadata required for later measurements;
- temporary sector-filtered operators used by direct products.

Block checkpoints may be written at every step or restricted to the latest state. Rotation matrices may remain cached in memory for speed or be stored on disk to reduce resident memory. Likewise, a measurement state can be saved so that local observables, entanglement quantities, and static correlations can be evaluated without repeating the expensive ground-state optimization.

The use of explicit quantum-number shifts for operators extends the same sector logic from the Hamiltonian to measurements. A correlation

$$\langle O_A(i) O_B(j) \rangle \quad (41)$$

is allowed within a fixed total sector only if

$$dq(O_A) + dq(O_B) = 0. \quad (42)$$

Operators residing on opposite sides of the cut are contracted with the same rectangular, symmetry-resolved tensor-product machinery used by the direct Hamiltonian application. Thus the measurement layer reuses the central mathematical abstraction rather than constructing full superblock operators.

11 End-to-end workflow

A complete calculation can be summarized as follows:

1. Define the local Hilbert space, local Hamiltonian, boundary operators, conserved Abelian quantum numbers, and operator shifts.
2. Initialize left and right blocks and choose the target total sector.
3. Enlarge both blocks and construct the symmetry-compatible superblock basis.
4. Select sparse, cached-direct, or lazy-direct Hamiltonian application.
5. Distribute every wavefunction sector matrix by columns and build sector-specific MPI communicators.
6. Solve the superblock eigenproblem through repeated distributed Hamiltonian–vector products.
7. Form the symmetry-block reduced density matrices, diagonalize them, and select the retained Schmidt states.
8. Renormalize the blocks and all required operators.
9. Repeat the growth process for infinite DMRG, or move the cut through the lattice during finite sweeps.
10. Store checkpoints, rotation matrices, and measurement states according to the selected memory policy.
11. Evaluate observables and correlations in post-processing when desired.

12 Scaling picture and practical regime

For sufficiently large sector matrices, local contractions dominate the cost, while the distributed transpositions introduce a subleading communication contribution. The ideal strong-scaling behavior is approximately

$$T(P) \sim \frac{T(1)}{P}, \quad (43)$$

until one of three effects becomes important:

- (i) symmetry fragmentation leaves too little work in small sectors;
- (ii) the local matrix panels approach cache- or latency-dominated sizes;
- (iii) filtering and collective-communication overhead becomes comparable with arithmetic.

Increasing the bond dimension generally improves parallel efficiency because it increases the local volume-to-communication-surface ratio. The relevant limit is consequently not only the formal complexity of DMRG, but also the entanglement- and symmetry-controlled distribution of the retained basis.

The approach is not conceptually restricted to DMRG. It requires a symmetry-preserving bipartition, a wavefunction represented in the associated product basis, and a Hamiltonian expressible as a manageable sum of products across the cut. DMRG provides a particularly natural realization because its Schmidt decomposition already constructs and truncates precisely this basis.

13 Summary

The central design principle of **nssDMRG** is that entanglement and symmetry organize both the physical approximation and the distributed computation. The entanglement cut produces a set of symmetry-resolved wavefunction matrices. These matrices define the retained DMRG basis, the MPI ownership layout, the allowed operator transitions, and the communication pattern of the Hamiltonian application.

The superblock Hamiltonian is reduced to local contractions of the form

$$BXA^\top, \tag{44}$$

with distributed transpositions used only when the contraction direction is incompatible with the current data layout. Symmetry-sector filtering removes forbidden work, while sector-specific communicators prevent fragmented blocks from involving idle processes. Cached and lazy modes expose an explicit time–memory trade-off without changing the underlying algorithm. Together with symmetry-aware truncation, periodic-growth strategies, restart support, and post-processing measurements, this yields a scalable DMRG framework in which computational organization follows directly from the quantum structure of the state.

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